Ridham Arora

Data Scientist / Data Analyst

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LinkedIn | GitHub | Portfolio

SUMMARY

Analyzing data with SQL, Python, Excel. Proficient knowledge in Statistics, Mathematics and other Analytics tools and technologies. Have worked on multiple end-to-end projects and love exploring, cleaning, and playing with real-world data to uncover insights.

SKILLS

Programming: Python , C++ , SQL , MySql
BI Tools: Power BI , MS-Excel
Data Science: Machine Learning , Data Analytics , Natural Language Processing , EDA , Data Mining
Libraries & Tools : Pandas , Numpy , Matplotlib , Seaborn , Scikit-learn , Jupyter , GitHub , Colab
Relevant Courses: Statistics , Probability , Database Management , Data Structure , Mathematics

EXPERIENCE

Jr. Data Scientist at, **Sep 2025 – Present**
Pranjali Grow Cap Pvt Ltd ,Delhi.

- Develop and implement **data-driven options trading strategies** for Indian (NSE/BSE/MCX) and US markets.
- Built an **incremental Python + Polars pipelin**e that cut dashboard load time on **100M–1B+ row datasets from hours to seconds** — only new data is re-processed.
- Created a **DuckDB-over-Parquet** query layer and **Streamlit dashboard** with an Excel-style drag-and-drop pivot builder and auto-generated **Excel heatmaps (xlwings)**
- Backtested strategies across **thousands of parameter combinations**, aggregating PnL by days-to-expiry, weekday and strike to surface top-performing setups.
- Engineered **memory-optimized ingestion** (RAR → Parquet, chunked aggregation) handling millions of rows per run under tight memory limits.

SELECTED PROJECTS

- **Biodiversity Analysis of U.S. National Parks:** Performed exploratory data analysis on endangered species using **pandas, NumPy, Matplotlib, Seaborn.** Identified conservation patterns across park categories. Delivered insights through interactive visualizations. *(Link)*
- **Car Price Prediction:** Predicted car prices using regression techniques from **CarDekho**. Built and evaluated models like **Random Forest Regressor, Ridge ,Lasso ,K-Nearest , Decision Tree** achieving $R^2 = 0.86$. Performed feature engineering and comparative model analysis. *(Link)*
- **Kindle Review Sentiment Analysis:** Built an NLP to classify customer sentiments using **Bag of Words, TF-IDF, and Naive Bayes.** Preprocessed text with NLTK, removed stopwords, and applied tokenization. Achieved normal accuracy in sentiment classification using Kindle reviews. *(Link)*

EDUCATION

Bachelor of Computer Applications **Oct ,2022 - June,2025**
Tecnia Institute of advanced studies , Delhi
Cumulative GPA: 7.9/10

Senior Secondary (12th), CBSE **Apr ,2021 - July ,2022**
Kendriya Vidyalaya no -3 ,Delhi
Percentage: 65%

CERTIFICATION

Machine learning I (Columbia+)**, Deloitte Data Analytics Job Simulation** (Forage)**, Getting Started with AI** (IBM SkillsBuild)**, Machine Learning for Data Science Projects** (IBM SkillsBuild)**,Journey to Cloud** (IBM SkillsBuild)